John Connolly

Principal Engineer | AI Engineer: LLM apps, RAG, evals, agents, MCP
Product & Forward-Deployed Engineering · Python, TypeScript, Azure, AWS, Terraform
john@connol.ly · linkedin.com/in/jeconnol · github.com/jconnolly · ollyconn.com

Experience

Data Society Group Remote
Lead Product Engineer (Principal-level IC) *Jan 2025 – Present*

- Ran engineering, delivery and product for a construction client's program while building the backend and running the infrastructure myself.
- **Project intake and financial operations platform:** reviewed the proposal and estimate early and shaped the schedule, roles and split of work across onshore, offshore and partner developers; took Figma designs to the data model and OpenAPI spec.
- Designed and built the integration with the client's ERP (contract and project creation, approval callbacks, outbox dispatch, reconciliation), with Playwright end-to-end tests confirming records landed in the ERP.
- Led delivery to go-live with the client and integration partner: Jira program, sprint planning, stakeholder reporting.
- **Generative AI document Q&A pilot (Azure):** Azure AI Document Intelligence ingestion into pgvector, hybrid version-aware retrieval, a **golden-question eval harness** with recorded baseline and audit trail; Terraform for Container Apps, GitHub OIDC deploys, Entra ID auth.
- AI-assisted development: parallel Claude Code agents in isolated devcontainers and git worktrees, adversarial PR review, verify-before-merge CI gates; documented for the team.
- **Internal / platform:** LLM teaching assistant (Bedrock + Claude, RAG over course content), AI course recommender, Post-Course Reporting app on ECS Fargate; owned Terraform and Bitbucket Pipelines; **dsg-tools**, an MCP server and Claude skills the team uses daily.

Career break – planned break for family time and hands-on work (boat restoration, car mechanics) *Feb 2024 – Dec 2024*

Flow Commerce (acquired by Global-e, NASDAQ: GLBE) Remote
Principal Software Engineer *Jul 2021 – Feb 2024*

- Tech lead for 3–5 engineers on the checkout, payments, tax and fulfillment engine behind **Shopify Markets Pro** (now Shopify Managed Markets). Scala, Akka, Play, PostgreSQL, AWS, Kubernetes.
- Took it from proof of concept to **$50M+ GMV** across **3,500+ merchants** and **136 international markets**; tuned for Shopify Plus Black Friday / Cyber Monday traffic.
- After the Global-e acquisition, merged Flow's stack into the joint product: catalog, currency, tax, order routing.

Bug Labs New York, NY
Lead Software Engineer (grew from QA to development lead, team of 7–9) *Feb 2008 – Apr 2012*

- SDK and rootfs for the **BUG** modular embedded Linux platform: OMAP kernels, Linaro/OpenEmbedded upstream work, drivers; Java/JNI/C APIs for hardware modules; XMPP/chunked-HTTP cloud messaging.
- DARPA street-level capture pilot (**OpenCV**, Python); Processing color-tracking demo at MoMA's *Talk to Me* (2011); IoT data service backend (now dweet.io).

Earlier Roles
Software Architect Team Lead, Netsmart (2020–2021) · Senior Applications Engineer, Perfumania (2018–2020) · Senior Software Engineer, Open-Xchange (2016–2017) · Senior Software Engineer, CooCoo (2015–2016) · Software Engineer, TheLadders (2012–2015)

Project

local-llm-pi5 2026

- Raspberry Pi 5 + Hailo-10H vs. Apple Silicon with measured tok/s. Hailo wins on compiled vision/Whisper models, not agentic LLMs (2B ceiling, 2k context).

Education & Publications

Stony Brook University – B.S. Computer Science *Stony Brook, NY*

- Grim, Selinger, Braynen, Rosenberger, Au, Louie, Connolly. "Reducing Prejudice: A Spatialized Game-Theoretic Model for the Contact Hypothesis." *Artificial Life IX*, MIT Press, 2004, pp. 244–250.
- Same authors. "Modeling Prejudice Reduction: Spatialized Game Theory and the Contact Hypothesis." *Public Affairs Quarterly* 19(2), 2005, pp. 95–125.

Skills

Leadership: Technical leadership, solutions architecture, stakeholder and program management, API design, data modeling
AI: Generative AI, RAG, LLM evals, agents, MCP, Claude API, Azure OpenAI, Azure Document Intelligence, Bedrock, pgvector
Engineering: Python, TypeScript, Scala, Terraform, Azure, AWS, Kubernetes, PostgreSQL, Playwright, CI/CD, Embedded Linux